

Homework 01 Solutions

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Course: 10-783 A
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Task 1.1

For an n -dimensional sample space (i.e., $x_0 \in \mathbb{R}^n$), we have that

$$\begin{aligned} W_2^2(\mathcal{N}(0, \mathbf{I}), \mathcal{N}(\sqrt{\bar{\alpha}_T} \mathbf{x}_0, (1 - \bar{\alpha}_T) \mathbf{I})) &= \|\sqrt{\bar{\alpha}_T} \mathbf{x}_0\|^2 + n(1 + (1 - \bar{\alpha}_T) - 2(1 - \bar{\alpha}_T)^{1/2}) \\ &= \bar{\alpha}_T \|\mathbf{x}_0\|^2 + n(2 - \bar{\alpha}_T - 2\sqrt{1 - \bar{\alpha}_T}). \end{aligned}$$

As $\bar{\alpha}_T \rightarrow 0$, the Wasserstein distance also approaches 0.

Task 1.2

We have

$$\begin{aligned} \log \bar{\alpha}_T &= \sum_{t=1}^T \log(1 - \beta_t) \geq -\sum_{t=1}^T \frac{\beta_t}{1 - \beta_t} \geq -\frac{1}{1 - \beta_{\max}} \sum_{t=1}^T \beta_t = -\frac{\frac{T}{2}(\beta_{\min} + \beta_{\max})}{1 - \beta_{\max}} \\ \therefore \bar{\alpha}_T &\geq \exp\left(-\frac{\frac{T}{2}(\beta_{\min} + \beta_{\max})}{1 - \beta_{\max}}\right) \end{aligned}$$

as the lower bound on $\bar{\alpha}_T$. The corresponding upper bound (using $\log(1 - x) \leq -x$) is

$$\log \bar{\alpha}_T \leq -\frac{T}{2}(\beta_{\min} + \beta_{\max}).$$

Together, this gives

$$-\frac{\frac{T}{2}(\beta_{\min} + \beta_{\max})}{1 - \beta_{\max}} \leq \log \bar{\alpha}_T \leq -\frac{T}{2}(\beta_{\min} + \beta_{\max}),$$

so $\log \bar{\alpha}_T = -\frac{T}{2}(\beta_{\min} + \beta_{\max})(1 + O(\beta_{\max}))$. If $\beta_{\max} \ll 1$, then the lower bound is approximately equal to the upper bound, since $1 - \beta_{\max} \approx 1$. For constant schedule $\beta' = \frac{1}{2}(\beta_{\min} + \beta_{\max})$, we have

$$\log \bar{\alpha}'_T = T \log(1 - \beta') = -T(\beta' + O(\beta'^2)) = -\frac{T}{2}(\beta_{\min} + \beta_{\max})(1 + O(\beta_{\max})).$$

This shows that the $\log \bar{\alpha}_T$ decays at the same rate in both schedules to the leading order of $\beta_{\max}$.

Task 1.3

This cosine scheduler is useful because for $1 \leq t \leq T$, $\bar{\alpha}_t$ approaches 0 slower compared to a linear scheduler for β_t (Figure 1). This means for fixed T that more steps are spent on denoising meaningful inputs that are not just pure noise. The change in $\bar{\alpha}_t$ is also relatively uniform w.r.t. T , which is necessary to ensure that $p(x_{t-1} | x_t)$ is near-Gaussian. Finally, the rates of change at $t/T \approx 0$ and $t/T \approx 1$ are small, which is useful in the former to ensure fine details in the input are not immediately noised out, and useful in the latter to keep β_t bounded as $\bar{\alpha}_t \rightarrow 0$.

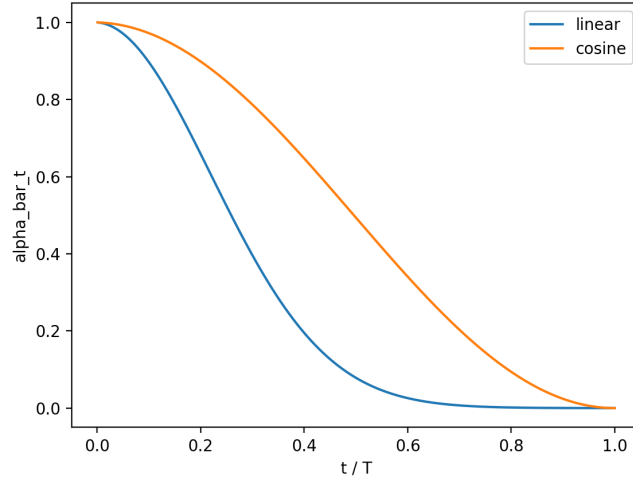


Figure 1: $\bar{\alpha}_t$ vs T for cosine and linear schedules. Parameters: $s = 0.08$, $T = 1000$, $\beta_{\min} = 10^{-4}$, $\beta_{\max} = 0.02$.

Task 2.1

(1) We first show that $\mathbf{P}\{|X_n - X| > \epsilon\} \leq \mathbf{E}[\epsilon^{-1} |X_n - X|]$:

$$\begin{aligned}
 \mathbf{E}[\epsilon^{-1} |X_n - X|] &= \int_{\Omega} \epsilon^{-1} |X_n - X| \, dP(\omega) \\
 &\geq \int_{\Omega} \epsilon^{-1} |X_n - X| I_{|X_n - X| > \epsilon} \, dP(\omega) \\
 &\geq \int_{\Omega} \epsilon^{-1} \cdot \epsilon I_{|X_n - X| > \epsilon} \, dP(\omega) \\
 &= \int_{\Omega} I_{|X_n - X| > \epsilon} \, dP(\omega) \\
 &= \mathbf{P}\{|X_n - X| > \epsilon\}.
 \end{aligned}$$

From Task 2.1.2, we have that $X_n \xrightarrow{L_2} X \Rightarrow X_n \xrightarrow{L_1} X$. For fixed $\epsilon > 0$, we have

$$\begin{aligned} 0 &= \lim_{n \rightarrow \infty} \mathbf{E}[|X_n - X|] = \lim_{n \rightarrow \infty} \mathbf{E}[\epsilon^{-1} |X_n - X|] \\ &= \limsup_{n \rightarrow \infty} \mathbf{E}[\epsilon^{-1} |X_n - X|] \\ &\geq \limsup_{n \rightarrow \infty} \mathbf{P}\{|X_n - X| > \epsilon\}. \end{aligned}$$

Since $\mathbf{P}\{\cdot\} \geq 0$, we have $\liminf_{n \rightarrow \infty} \mathbf{P}\{|X_n - X| > \epsilon\} = 0$, showing that

$$\lim_{n \rightarrow \infty} \mathbf{P}\{|X_n - X| > \epsilon\}$$

exists and is equal to 0 by the Squeeze theorem, giving $X_n \xrightarrow{p} X$, as desired.

- (2) From Hölder's inequality with $a = p/q$, $b = \frac{p}{p-q}$ (satisfying $a, b > 1$ since $p > q$ and $1/a + 1/b = 1$ by construction), we get

$$\mathbf{E}[|X_n - X|^q \cdot 1] \leq \mathbf{E}[|X_n - X|^{q \cdot a}]^{1/a} \cdot (1^b)^{1/b} = \mathbf{E}[|X_n - X|^p]^{q/p}.$$

Since $X_n \xrightarrow{L_p} X$, we have that

$$0 = \lim_{n \rightarrow \infty} \mathbf{E}|X_n - X|^p = \lim_{n \rightarrow \infty} \mathbf{E}[|X_n - X|^p]^{q/p}.$$

Together, this shows that

$$\lim_{n \rightarrow \infty} \mathbf{E}|X_n - X|^q \leq \lim_{n \rightarrow \infty} \mathbf{E}[|X_n - X|^p]^{q/p} = 0.$$

Since $|X_n - X| \geq 0$, the limit exists and is equal to 0.

- (3) The implication $X_n \xrightarrow{L_2} X \Rightarrow X_n \xrightarrow{L_1} X$ follows immediately from Task 2.1.2 and the implication $X_n \xrightarrow{p} X \Rightarrow X_n \xrightarrow{d} X$ is given. For the middle implication, we showed in Task 2.1.1 that the stronger statement of $X_n \xrightarrow{L_1} X \Rightarrow X_n \xrightarrow{p} X$ holds. Together, this shows that the implication chain holds at each step.

Task 2.2

Let $I_{[a,b]}$ denote the indicator variable that is 1 on $[a, b]$ and 0 elsewhere.

- (1) Consider $X_n = n^{1/2} I_{[0, 1/n]}$ and $X = 0$ defined on $U(0, 1)$. We have that

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbf{E} \left| n^{1/2} I_{[0, 1/n]} \right|^2 &= \lim_{n \rightarrow \infty} 1/n \cdot n = 1 \\ \lim_{n \rightarrow \infty} \mathbf{E} \left| n^{1/2} I_{[0, 1/n]} \right|^1 &= \lim_{n \rightarrow \infty} 1/n \cdot n^{1/2} = 0, \end{aligned}$$

showing that $X_n \xrightarrow{L_1} X$ but not $X_n \xrightarrow{L_2} X$.

(2) Consider $X_n = nI_{[0,1/n]}$ and $X = 0$ defined on $U(0,1)$. We have that

$$\lim_{n \rightarrow \infty} \mathbf{E} |nI_{[0,1/n]}|^1 = \lim_{n \rightarrow \infty} 1/n \cdot n = 1,$$

so X_n is not L_1 convergent to X . Then,

$$\lim_{n \rightarrow \infty} \mathbf{P} \{|X_n - X| > \epsilon\} = \lim_{n \rightarrow \infty} \mathbf{P} \{X_n > \epsilon\} = \lim_{n \rightarrow \infty} \mathbf{P} \{U(0,1) \leq 1/n\} = \lim_{n \rightarrow \infty} 1/n = 0,$$

showing that $X_n \xrightarrow{p} X$.

(3) Let $X = U(\{-1,1\})$ and $X_n = -X$ for all $n \geq 1$. Since X and X_n have the same distribution by symmetry, $X_n \xrightarrow{d} X$ trivially. However, $|X_n - X| = 2$ for all n , so X_n does not converge to X in probability.

Task 2.3

For a bounded continuous statistic f , if $P_n \xrightarrow{d} P_{\text{data}}$, by the Portmanteau theorem it is guaranteed that

$$\mathbf{E}_{X \sim P_n} f(X) \rightarrow \mathbf{E}_{X \sim P_{\text{data}}} f(X),$$

or in other words, the statistic on the generated distribution approaches the statistic on the underlying data distribution. This means that MMD converges if the sequence of generative models approaches the data distribution. This is because

$$\text{MMD}^2(P_n, P_{\text{data}}) = \mathbf{E}_{X, X' \sim P_n} [k(X, X')] + \mathbf{E}_{Y, Y' \sim P_{\text{data}}} [k(Y, Y')] - 2\mathbf{E}_{X \sim P_n, Y \sim P_{\text{data}}} [k(X, Y)],$$

where $X, X' \sim P_n$ and $Y, Y' \sim P_{\text{data}}$ are independent. For a bounded continuous kernel k , each term is an expectation of a bounded continuous function and thus converges under weak convergence, giving $\text{MMD}(P_n, P_{\text{data}}) \rightarrow 0$ as $P_n \xrightarrow{d} P_{\text{data}}$. On the other hand, an unbounded statistic like FID has no such guarantees.

Task 3.1

We have that

$$\begin{aligned} x_t &= \sqrt{\alpha_t} \left(\sqrt{\bar{\alpha}_{t-1}} x_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon' \right) + \sqrt{\beta_t} z_t \\ &= \sqrt{\bar{\alpha}_{t-1} \alpha_t} x_0 + \sqrt{(1 - \bar{\alpha}_{t-1}) \alpha_t} \epsilon' + \sqrt{\beta_t} z_t \\ &= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{(1 - \bar{\alpha}_{t-1}) \alpha_t + \beta_t} \epsilon \\ &= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{\alpha_t - \bar{\alpha}_t + \beta_t} \epsilon \\ &= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \end{aligned}$$

where $\epsilon \sim \mathcal{N}(0, I)$, showing the induction hypothesis. In particular, step 2 to 3 follows from the additivity of two independent Gaussians. Then,

$$\mathbf{E}[x_t | x_0] = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \mathbf{E}[\epsilon] = \sqrt{\bar{\alpha}_t} x_0,$$

as desired.

Task 3.2

We have that

$$\text{Cov}(x_t | x_0) = \mathbf{E} [(x_t - \mu)(x_t - \mu)^\top | x_0],$$

where $\mu = \mathbf{E}[x_t | x_0] = \sqrt{\bar{\alpha}_t}x_0$. Then, $x_t - \mu = \sqrt{1 - \bar{\alpha}_t}\epsilon$. This gives

$$\mathbf{E} [(x_t - \mu)(x_t - \mu)^\top | x_0] = \mathbf{E} [(\sqrt{1 - \bar{\alpha}_t}\epsilon)(\sqrt{1 - \bar{\alpha}_t}\epsilon)^\top | x_0] = (1 - \bar{\alpha}_t)\mathbf{E} [\epsilon\epsilon^\top] = (1 - \bar{\alpha}_t)\mathbf{I}.$$

Since each Markov step adds independent Gaussian noise, the distribution of x_t is an affine function of x_0 plus a sum of independent Gaussians, so conditioned on x_0 it is Gaussian. As shown above, the distribution has mean $\sqrt{\bar{\alpha}_t}x_0$ and covariance $(1 - \bar{\alpha}_t)\mathbf{I}$, giving

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathbf{I}),$$

as desired.

Task 3.3

We have

$$\begin{aligned} \mathbf{E}\|x_t\|^2 &= \mathbf{E} [(\sqrt{\bar{\alpha}_t}x_0)^\top (\sqrt{\bar{\alpha}_t}x_0) + 2(\sqrt{\bar{\alpha}_t}x_0)^\top (\sqrt{1 - \bar{\alpha}_t}\epsilon) + (\sqrt{1 - \bar{\alpha}_t}\epsilon)^\top (\sqrt{1 - \bar{\alpha}_t}\epsilon)] \\ &= \bar{\alpha}_t \mathbf{E} [x_0^\top x_0] + (1 - \bar{\alpha}_t) \mathbf{E} [\epsilon^\top \epsilon] \\ &= \bar{\alpha}_t d + (1 - \bar{\alpha}_t) d \\ &= d. \end{aligned}$$

The removal of the cross term is due to the independence of x_0 and ϵ . As $\bar{\alpha}_T \rightarrow 0$, $q(x_T | x_0)$ approaches $\mathcal{N}(0, \mathbf{I})$. For the given formula for β_t , $\bar{\alpha}_{100} \approx 0.602$, $\sqrt{\bar{\alpha}_{100}} \approx 0.776$, $\sqrt{1 - \bar{\alpha}_{100}} \approx 0.630$.

Task 4.1

We have

$$\text{Var}(W_t - W_s) = \text{Cov}(W_t - W_s, W_t - W_s) = \text{Var}(W_t) + \text{Var}(W_s) - 2\text{Cov}(W_s, W_t) = t + s - 2s = t - s.$$

This is consistent with the distribution of $W_t - W_s$ being $\mathcal{N}(0, t - s)$.

Task 4.2

We have

$$\begin{aligned} \text{Var}(W_1 + W_2) &= \text{Var}(2W_1 + (W_2 - W_1)) = 4\text{Var}(W_1) + \text{Var}(W_2 - W_1) = 5 \\ \text{Cov}(W_1, W_3 - W_2) &= 0. \end{aligned}$$

The intervals $[0, 1]$ and $[2, 3]$ are disjoint, so the Wiener process increments on those intervals are independent, implying a covariance of 0.

Task 4.3

We have

$$\text{Corr}(W_s, W_t) = \frac{\text{Cov}(W_s, W_t)}{\sqrt{\text{Var}(W_s) \text{Var}(W_t)}} = \frac{s}{\sqrt{st}} = \sqrt{s/t}.$$

For $s = 1, t = 4$, $\text{Corr}(W_s, W_t) = 1/2$. As $t \rightarrow \infty$ with s fixed, $\text{Corr}(W_s, W_t) \rightarrow 0$. This shows that earlier values for a Wiener process matter increasingly less as the time window grows, since $\text{Var}(W_t)$ grows without bound and W_s remains constant.

Task 5.1

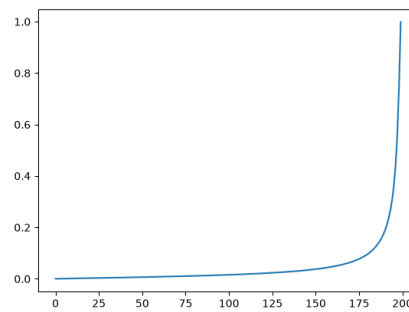


Figure 2: β_t for a cosine schedule for $t \in [0, 200]$

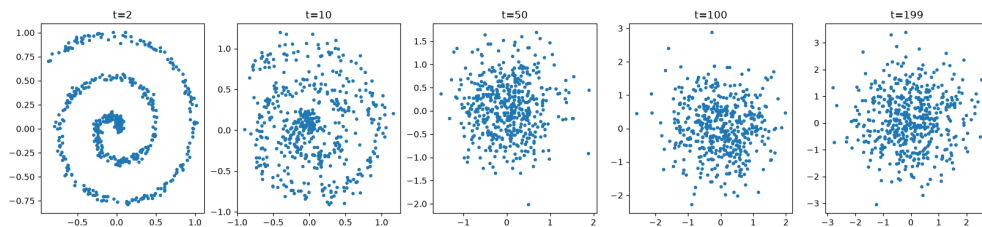


Figure 3: Forward diffusion example

Collaboration Questions

- (1) (a) Did you receive any help whatsoever from anyone in solving this assignment? No.
(b) If you answered 'yes', give full details. N/A.
- (2) (a) Did you give any help whatsoever to anyone in solving this assignment? No.
(b) If you answered 'yes', give full details. N/A.
- (3) (a) Did you find or come across code that implements any part of this assignment? No.
(b) If you answered 'yes', give full details. N/A.

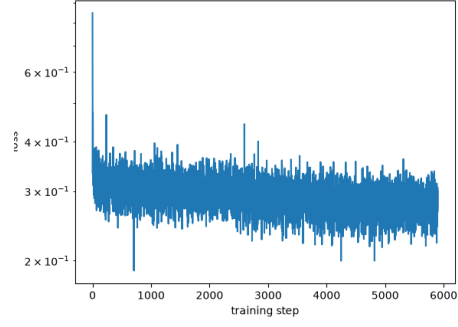


Figure 4: MSE loss of a diffusion model

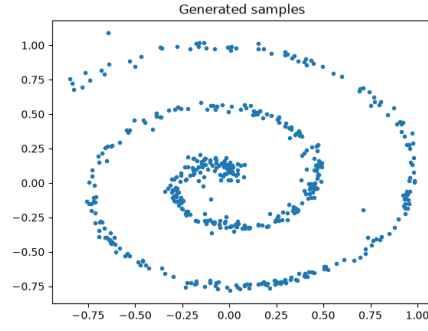


Figure 5: Generated samples of reverse diffusion for $n = 500$ points

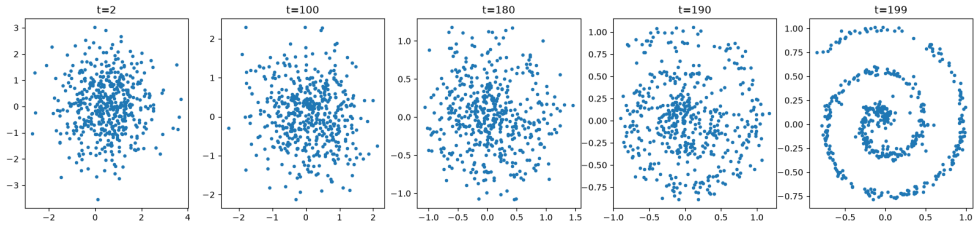


Figure 6: Reverse diffusion process for $n = 500$ points